

Milad Alipour Shahraki

📍 Boston MA, USA | ✉ alipourshahraki.m@northeastern.edu | 🏠 miladalipourshahraki.github.io | 📄 Google Scholar

SUMMARY

Ph.D. candidate in control theory (nonlinear, optimal, and safety-critical control). My work combines approximate-HJB optimal control with control-barrier-function safety constraints for real-time control of nonlinear systems, with applications in aerospace and autonomous systems. I implement in Python, Julia, and MATLAB/Simulink.

EDUCATION

- Ph.D. in Electrical Engineering | Northeastern University, Boston, MA** Sep. 2022 – Present
Dissertation: Safe Near-Optimal Control of Nonlinear Systems
Advisor: Prof. Laurent Lessard
CGPA: 4.00 / 4.00
- M.Sc. in Mechanical Engineering | University of Isfahan, Isfahan** Sep. 2017 – Jan. 2020
Thesis: Fractional-Order Sliding Mode Control of Flexible Spacecraft
Advisor: Prof. Maryam Malekzadeh • Co-Advisor: Prof. Alireza Ariaei
CGPA: 3.88 / 4.00 (Ranked 1st in class)
- B.Sc. in Mechanical Engineering | Fasa University, Fars** Sep. 2013 – Jul. 2017
Thesis: Control Under External Disturbance Using Disturbance Observers
Advisor: Prof. Mohamadreza Homayounzade
CGPA: 3.59 / 4.00 (Ranked 1st in class)

RESEARCH EXPERIENCE

- Research Assistant | Lessard's Research Group, Northeastern University** Sep. 2022 – Present
- Developed a two-stage near-optimal control framework: offline approximate-HJB value function (SOS / policy iteration) plus an online safety-constrained QP
 - Matched open-loop optimal performance on benchmark problems at real-time solve rates, with safe-set constraints modifiable online without recomputing the value function, decoupling performance optimization from constraint enforcement
 - Designed real-time CLF/CBF-QP controllers for agile spacecraft attitude control under reaction-wheel torque and angular-momentum constraints; enforced state/input limits as hard constraints and eliminated CLF-QP input chattering at low (10 Hz) sampling rates
 - Benchmarked against MPC, constrained LQR, RES-/OD-CLF-CBF-QP, and open-loop optimal control in Julia (JuMP/HiGHS/Ipopt) and CVXPY/PROXQP
- Research Assistant | Mechatronics Lab, University of Isfahan** Sep. 2017 – Jan. 2020
- Designed adaptive fractional-order nonsingular terminal sliding-mode controllers for rigid and flexible spacecraft attitude control under actuator saturation and disturbances, with reaction-wheel momentum management
 - Proved practical fixed-time stability via Lyapunov analysis and validated experimentally on a spacecraft attitude-control simulator (LabVIEW, four reaction wheels on Maxon DC motors, AHRS feedback, 40 ms sampling), driving reaction-wheel momentum to rest after each maneuver
- Research Assistant | Homayounzade's Research Group, Fasa University** Feb. 2016 – Jul. 2017
- Developed an adaptive output-feedback trajectory-tracking controller for unmanned surface vessels with a velocity observer and disturbance estimator, requiring only position and orientation measurements; proved asymptotic tracking via Lyapunov/Barbalat analysis

PUBLICATIONS

Journal Papers

- Milad Alipour Shahraki, Maryam Malekzadeh, and Alireza Ariaei, "Practical fractional-order nonsingular terminal sliding mode control of spacecraft," *ISA Transactions*, vol. 128, pp. 162–173, 2022. DOI: [10.1016/j.isatra.2021.10.022](https://doi.org/10.1016/j.isatra.2021.10.022)
- Milad Alipour Shahraki, Maryam Malekzadeh, and Alireza Ariaei, "Active fractional-order sliding mode control of flexible spacecraft under actuators saturation," *Journal of Sound and Vibration*, vol. 535, 117110, 2022. DOI: [10.1016/j.jsv.2022.117110](https://doi.org/10.1016/j.jsv.2022.117110)
- Mohamadreza Homayounzade and Milad Alipour Shahraki, "Output feedback adaptive control of dynamically positioned surface vessels: a disturbance observer-based control approach," *International Journal of Robotics and*

Conference Papers

4. **Milad Alipour Shahraki** and Laurent Lessard, “Near-Optimal Constrained Feedback Control of Nonlinear Systems via Approximate HJB and Control Barrier Functions,” *Accepted, 2026 IEEE Conference on Decision and Control (CDC)*, 2026. DOI: 10.48550/arXiv.2603.16114
5. **Milad Alipour Shahraki** and Laurent Lessard, “Spacecraft Attitude Control Under Reaction Wheel Constraints Using Control Lyapunov and Control Barrier Functions,” *2025 American Control Conference (ACC)*, pp. 947–952, IEEE, 2025. DOI: 10.23919/ACC63710.2025.11108025

SELECTED COURSE PROJECTS

Multi-Agent Target Localization and Safe Navigation | Safe Control and Estimation (ME7374)

Fall 2024

Optimal agent-target geometry via Fisher-Information (D-optimality) maximization; decentralized information-form EKF for cooperative estimation, with an EKF disturbance observer, Lyapunov-based tracking, and a CBF-QP safety filter for collision avoidance

MPC for Aircraft under STL Specifications | Formal Methods for Dynamical Systems (EECE5698)

Spring 2023

Encoded STL specifications as mixed-integer linear constraints; synthesized control via receding-horizon MILP (YALMIP) for a linearized aircraft, comparing one-shot, MPC, and pseudo-MPC on feasibility and solve time

SKILLS

Control and Optimization

Model Predictive Control (MPC) • Control Barrier / Lyapunov Functions (CBF/CLF) • HJB / approximate dynamic programming • convex optimization / real-time quadratic programming (QP) • sum-of-squares (SOS) • signal temporal logic (STL) / mixed-integer programming • sliding-mode / adaptive control • fixed-time stability analysis • multi-agent / decentralized control • state estimation and observer design • hardware-in-the-loop validation

Software

Python • Julia • MATLAB and Simulink • LabVIEW • $\text{\LaTeX}$ • C++ • solvers/frameworks: JuMP, CVXPY, MOSEK, Ipopt, PROXQP, HiGHS, YALMIP

Other

Computer vision (visual odometry, feature detection/tracking, pose estimation) • SolidWorks • AutoCAD

Languages

English (Fluent) • Persian (Native)

PROFESSIONAL ACTIVITIES AND SERVICE

Journal Reviewer

IEEE Transactions on Automatic Control • IEEE Transactions on Systems, Man, and Cybernetics: Systems • IEEE/ASME Transactions on Mechatronics • IEEE Control Systems Letters • Automatica • ISA Transactions • Aerospace Science and Technology • Control Engineering Practice

Conference Reviewer

Learning for Dynamics and Control Conference (L4DC) • IEEE Conference on Decision and Control (CDC)

AWARDS AND HONORS

University of Isfahan

Ranked 1st in Mechanical Engineering • *Exceptional Talent Student* scholarship (all semesters)

Fasa University

Ranked 1st in Mechanical Engineering • *Exceptional Talent Student* scholarship (all semesters)

TEACHING EXPERIENCE

Teaching Assistant | Northeastern University

2025 – 2026

Mathematical Methods for Mechanical Engineers (ME6200) • Fundamentals of Linear Systems (EECE2520) • Statics (ME2350)

Teaching Assistant | Fasa University

2015 – 2016

Statics • Engineering Drawing and Sketching

REFERENCES

Available upon request.